

DSP EEE/ECE F434

Tut 10

1. Frequency transformations for an analog LPF to an LPF, HPF, BPF, and BSF.

Type of transformation	Transformation
Lowpass	$s \rightarrow \frac{\Omega_p}{\Omega'_p} s$
Highpass	$s \rightarrow \frac{\Omega_p \Omega'_p}{s}$
Bandpass	$s \rightarrow \Omega_p \frac{s^2 + \Omega_l \Omega_u}{s(\Omega_u - \Omega_l)}$
Bandstop	$s \rightarrow \Omega_p \frac{s(\Omega_u - \Omega_l)}{s^2 + \Omega_u \Omega_l}$

2.

Show that the analog transfer function

$$H_1(s) = \frac{bs}{s^2 + bs + \Omega_o^2}, \quad b > 0,$$

has a bandpass magnitude response with $|H_a(j0)| = |H_a(j\infty)| = 0$ and $|H_a(j\Omega_o)| = 1$. Determine the frequencies Ω_1 and Ω_2 at which the gain is 3 dB below the maximum value of 0 dB at Ω_o . Show that $\Omega_1 \Omega_2 = \Omega_o^2$. The difference $\Omega_2 - \Omega_1$ is called the 3-dB bandwidth of the bandpass transfer function. Show that $b = \Omega_2 - \Omega_1$.

3.

Show that the analog transfer function

$$H_2(s) = \frac{s^2 + \Omega_o^2}{s^2 + bs + \Omega_o^2}, \quad b > 0,$$

has a bandstop magnitude response with $|H_a(j0)| = |H_a(j\infty)| = 1$ and $|H_a(j\Omega_o)| = 0$. Since the magnitude is exactly zero at Ω_o , it is called the notch frequency, and $H_2(s)$ is often called the notch transfer function. Determine the frequencies Ω_1 and Ω_2 at which the gain is 3 dB below the maximum value of 0 dB at $\Omega = 0$ and $\Omega = \infty$. Show that $\Omega_1 \Omega_2 = \Omega_o^2$. The difference $\Omega_2 - \Omega_1$ is called the 3-dB notch bandwidth of the bandpass transfer function. Show that $b = \Omega_2 - \Omega_1$.

4.

The bandpass transfer function $H_1(s)$ and the bandstop transfer function $H_2(s)$ can be expressed in the form

$$H_1(s) = \frac{1}{2}[A_1(s) - A_2(s)], \quad H_2(s) = \frac{1}{2}[A_1(s) + A_2(s)],$$

where $A_1(s)$ and $A_2(s)$ are stable analog allpass transfer functions. Determine $A_1(s)$ and $A_2(s)$.

Note: $H_1(s)$ and $H_2(s)$ are of Q2 and Q3